UPI & Online *Payment Fraud*

A detailed case study — what it is, how it works

CRIME TYPE:
UPI / Online Payment Fraud

CLASSIFICATION
Social Engineering + Financial Cybercrime

JURISDICTION
India (I4C / NCIB / RBI Data)

SCENARIO TYPE
Real Published Data + Fictional Scenario

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ABSTRACT

This case study examines UPI (Unified Payments Interface) fraud — a rapidly escalating form of digital financial crime in India. Using a combination of real statistical data from government and regulatory bodies and a realistic fictional scenario, this document details the nature of the crime, the precise psychological and technical steps attackers follow, the demographic groups most at risk, and the cascading personal, legal, and societal consequences that follow. Six primary sources are cited and described in full in the final section.

SECTION 01

What Is UPI / Online Payment Fraud?

Unified Payments Interface (UPI) is a real-time interbank payment system developed by the National Payments Corporation of India (NPCI) and launched in 2016. By 2023-24, UPI was processing over 13 billion transactions per month, making it one of the largest and most actively used digital payment ecosystems in the world. Its simplicity — linking a bank account to a mobile number and a six-digit PIN — made it accessible to hundreds of millions of Indians, including many who were entirely new to digital finance.

UPI fraud is a category of financial cybercrime in which malicious actors exploit the architecture, psychology, and user inexperience of the UPI ecosystem to trick individuals into authorising fund transfers to fraudulent accounts. Unlike malware-based banking attacks that require technical exploitation, the vast majority of UPI fraud is powered almost entirely by social engineering — the manipulation of human trust, fear, and urgency rather than software vulnerabilities.

The critical deception at the heart of UPI fraud is the confusion between a collect request and a payment receipt. UPI allows any registered user to send a collect request — essentially a demand for money — to another user's Virtual Payment Address (VPA). When the recipient enters their UPI PIN to confirm, they are authorising an outgoing debit, not receiving an incoming credit. Fraudsters exploit this distinction deliberately: they instruct victims to 'approve a refund' or 'accept a cashback' when in fact the PIN entry sends money out of the victim's account.

The crime operates across multiple vectors — phone calls, SMS phishing links, fake QR codes, fraudulent customer-care numbers listed on Google, job offer platforms, matrimonial websites, and social media marketplaces. It has evolved significantly in sophistication since 2018, now incorporating caller ID spoofing, data-breach exploitation, scripted psychological manipulation, and organised mule account networks.

SECTION 02

How It Typically Happens — Step by Step

While UPI fraud manifests across many pretexts and platforms, the underlying attack sequence follows a consistent, well-documented pattern. The following breakdown covers the canonical refund scam variant — the most widely reported form — in full detail.

1

Step 1 — Reconnaissance and Target Identification

The attacker begins by identifying a target. This may be opportunistic — scraping OLX, Quikr, or Facebook Marketplace for people actively selling goods, who are already expecting payment. It may also be data-driven: fraudsters purchase datasets from underground markets containing names, phone numbers, partial bank account details, or recent transaction records leaked from breached third-party apps. Targets are also identified through consumer grievance forums, where a public complaint telegraphs both identity and frustration — making the person ripe for a fake 'resolution' call.

2

Step 2 — Initial Contact and Authority Establishment

The attacker contacts the victim by phone, posing as a bank customer care executive, an e-commerce refund agent, a government official, or a buyer interested in a listed item. They use caller ID spoofing to make the number appear legitimate. Within the first 30 seconds, they cite verifiable personal details — the victim's full name, city, partial account number, or last transaction amount — sourced from breach data. This 'proof of knowledge' establishes false credibility rapidly and neutralises the victim's initial scepticism.

3

Step 3 — Creating the Pretext: Urgency and Opportunity

The attacker presents a scenario requiring immediate action. Common pretexts include: 'Your account has been flagged for suspicious activity and will be blocked within 2 hours unless you verify now'; 'We are processing a refund of Rs. 12,000 — please accept it within 10 minutes before the session expires'; or 'You have been selected for a government subsidy disbursement; verify your UPI to receive the funds immediately.' The artificial urgency prevents the victim from pausing to independently verify the claim or consult a family member.

4

Step 4 — Sending the Collect Request

While keeping the victim on the line, the attacker opens their UPI app and sends a collect request to the victim's UPI ID for a specific amount — for example, Rs. 9,999. The victim's phone buzzes with a notification. The attacker instructs: 'You will see a notification now — that is the refund arriving. Please enter your UPI PIN to accept it.' The phrase 'enter your PIN to accept' is the pivotal lie. The UPI interface itself shows an 'Enter PIN to pay' button, but in the stress of the call with an apparent 'bank official,' most victims do not process this language critically.

5

Step 5 — The Transaction: Instant and Irreversible

The moment the victim enters their UPI PIN and confirms, the transaction completes in under four seconds. The amount leaves the victim's account and lands in a fraudster-controlled mule account — typically a newly opened or borrowed account with minimal KYC, registered under a different name in a different city. UPI transactions are, by design, final and instantaneous. There is no 'pending' window, no fraud-hold mechanism, and no recall option. The attacker confirms receipt on their screen, thanks the victim warmly, promises the refund 'will reflect within two hours,' and disconnects.

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Step 6 — Fund Layering and Dispersal

Within minutes, the funds in the mule account are moved further — split into smaller amounts and pushed across a chain of two to four secondary mule accounts, or immediately withdrawn as cash from ATMs in different cities. Organised fraud rings manage networks of hired mule accounts remotely, making the money trail extremely difficult for law enforcement to follow across state lines. By the time the victim realises what has happened and contacts their bank, the funds are typically fragmented, partially withdrawn, and in some cases already converted to cryptocurrency or gift vouchers.

SECTION 02-A

Illustrative Fictional Scenario

The following scenario is a realistic composite based on documented fraud patterns, police FIR records, and consumer complaint data. All names are fictional and invented for illustrative purposes only.

Ramesh Gupta, 58, a retired schoolteacher from Bhopal, listed his old motorcycle on OLX for Rs. 35,000 in March 2024. Within a day, he received a call from someone identifying themselves as 'Sanjay Verma from PhonePe Customer Support.' The caller knew Ramesh's name, his OLX username, and the motorcycle's listed price — all publicly available on the platform.

'Sanjay' explained that an army officer stationed in Assam wished to buy the motorcycle but could only make payment through a special UPI verification process for defence personnel. He would be sending Rs. 2,000 as a token advance first. Ramesh received a UPI collect notification on his phone for Rs. 2,000. 'Just enter your PIN to receive it,' said the voice on the phone, cheerful and professional.

Ramesh entered his PIN. Within seconds, Rs. 2,000 left his account. Before he could process this, 'Sanjay' said the system required him to repeat the process for the full amount of Rs. 35,000 — described as a 'two-step defence verification.' Confused but still trusting the authority of the caller, Ramesh entered his PIN a second time. Rs. 35,000 was gone in under four seconds.

When Ramesh called his bank, he was told both transactions were authorised by him and were non-reversible. His complaint to the National Cybercrime Reporting Portal (cybercrime.gov.in) resulted in an FIR being registered, but the funds were traced to a mule account in West Bengal that had already been emptied before investigators could act on the freeze request.

[Fictional scenario — realistic composite based on fraud patterns documented in the RBI Annual Report 2023-24 and I4C Annual Report 2023. Names and all personal details are entirely invented for illustrative purposes.]

SECTION 02-B

Scale of the Problem — Key Statistics

1.12 Lakh+
UPI-related fraud complaints received on the National Cybercrime Reporting Portal in 2023 alone (I4C Annual Report 2023)

Rs. 1,700 Cr+
Estimated total financial lossesfrom UPI fraud in 2023 (MHA / I4C data)

67%
Shareof all digital payment fraud complaints that are UPI-related (RBI Annual Report 2023-24)

< 4 sec
Timefora UPI transaction tocomplete — leaving virtually no window to reverse the decision

SECTION 03

Who Is Usually Targeted?

UPI fraud does not target randomly. Certain demographic and behavioural profiles are significantly over-represented in victim data compiled by I4C, state cybercrime cells, and consumer protection organisations. Understanding these profiles is essential for designing effective awareness and prevention interventions.

Elderly and Semi-Urban Users

Individuals over 50, especially those who adopted UPI during or after COVID-19, often lack deep familiarity with the difference between sending and receiving money on the platform. They are also more trusting of authoritative-sounding callers. Rural and semi-urban users accessing UPI through feature phones or shared devices face additional barriers to identifying on-screen fraud indicators.

Online Marketplace Sellers

People listing items on OLX, Quikr, or Facebook Marketplace are already expecting payment and are psychologically primed to receive a collect request. Fraudsters scrape these platforms specifically for recently posted listings and call within hours, exploiting the seller's eagerness to complete the transaction quickly.

Job Seekers and Gig Workers

Fraudsters pose as HR representatives offering part-time work with UPI payment. Victims are asked to 'register' by making a small payment that will supposedly be refunded with their first week's earnings. Young adults actively searching for income are especially vulnerable; this variant circulates heavily on Telegram and WhatsApp.

People with Recent Consumer Complaints

Anyone who has publicly filed a complaint on a brand's social media page or forum is broadcasting their identity and frustration. Fraudsters monitor these posts and call posing as the brand's escalation team, offering a 'resolution' that requires the victim to enter a PIN to receive a refund — exploiting the victim's pre-existing desire for a quick resolution.

First-Generation UPI Adopters

Users who adopted UPI recently — often guided through setup by someone else without formal digital literacy training — frequently do not understand what their PIN authorises. They may genuinely believe the PIN is a simple identity verification step rather than a payment authorisation, which is precisely the misunderstanding the fraud exploits.

Matrimonial and Romance Contexts

A growing variant targets matrimonial app users. After weeks of cultivated trust, the fraudster fabricates a financial emergency — medical bills, a flight ticket, a customs payment for a gift. The emotional investment makes victims far more willing to send money and less likely to apply rational scrutiny. This variant often involves much larger amounts and repeated transactions over time.

SECTION 04

Consequences of UPI Fraud

The harm caused by UPI fraud extends well beyond the immediate financial loss. It radiates outward — affecting the victim's mental health, family relationships, and trust in digital systems, while also placing mounting pressure on law enforcement, the judiciary, and the broader digital payment ecosystem.

Consequences for the Victim

FINANCIAL

Immediate and largely unrecoverable financial loss. RBI data indicates fewer than 25% of UPI fraud complaints result in any recovery, and full recovery is far rarer. For many victims — retired individuals, daily wage earners, small traders — the lost amount represents weeks or months of income, emergency savings, or funds set aside for a child's education or wedding. The irreversibility of UPI transactions means that even a timely complaint to the bank cannot recall funds once they have been layered through mule accounts and partially withdrawn.

EMOTIONAL

Profound shame, guilt, and self-blame. Victims frequently describe feeling that they 'should have known better,' a sentiment often compounded when family members react with frustration or disbelief. Older victims in particular may develop lasting distrust of all digital financial tools, effectively self-excluding from the digital economy. Anxiety, persistent sleep disturbance, and clinical depression have been documented in victims who lost large sums — particularly when the funds were borrowed or represented long-term savings accumulated over years.

LEGAL

Difficult and slow complaint processes for victims. Filing a complaint involves navigating the National Cybercrime Reporting Portal (cybercrime.gov.in) or visiting a local cybercrime cell — processes many first-generation digital users find intimidating or physically inaccessible. Even after an FIR is registered, investigations are slow due to multi-state mule account chains and jurisdictional fragmentation between state police forces. Victims often wait months or years for resolution, creating secondary erosion of trust in institutions.

SYSTEMIC

Macro-level cost to digital inclusion and the payment ecosystem. The volume of UPI fraud imposes significant reputational and compliance costs on banks and NPCI, and erodes public confidence in UPI adoption — particularly in Tier 2 and Tier 3 cities where growth is still in progress. Law enforcement agencies face a rapidly growing caseload with low prosecution rates. The I4C 2023 data showing 1.12 lakh complaints translates to approximately 300 new fraud victims every single day.

Consequences for Perpetrators (If Caught)

Perpetrators, when identified and arrested, face charges under multiple legal frameworks. The Information Technology Act, 2000 — Section 66C (identity theft) and Section 66D (cheating by personation using a computer resource) — provides for imprisonment of up to three years and fines up to Rs. 1 lakh per offence. Additionally, IPC Section 420 (cheating) and Section 406 (criminal breach of trust) carry potential imprisonment of up to seven years.

Mule account holders — individuals who lend or sell their bank accounts to fraud rings — are increasingly prosecuted as co-conspirators rather than innocent facilitators. The Reserve Bank of India and NPCI have deployed automated mule account detection algorithms, and banks that fail to flag suspicious mule activity face regulatory consequences under RBI's master circular on fraud risk management. Prosecution rates, however, remain low: a significant share of reported cases are closed due to untraceable perpetrators, incomplete evidence, or account holders who successfully claim ignorance of the fraud ring.

SECTION 05

Sources

All sources cited are publicly available official documents, regulatory publications, or government reports. Each is described in full below — including its content, scope, and the specific relevance to this case study.

[1] ***Indian Cyber Crime Coordination Centre (I4C) — Annual Report 2023***

Published by the Ministry of Home Affairs (MHA), Government of India, the I4C Annual Report 2023 is the primary national compilation of cybercrime statistics and trends in India. It draws on complaint data from the National Cybercrime Reporting Portal (cybercrime.gov.in), FIR data from state cybercrime police units, and internal intelligence from the Cyber Fraud Mitigation Centre (CFMC). The report documents over 1.12 lakh UPI-related fraud complaints for 2023 and estimates total digital fraud losses — the majority UPI-driven — at over Rs. 1,700 crore. It provides breakdowns by fraud type, victim demographics, state-wise distribution, and modus operandi. Available on the I4C official portal and the MHA website (mha.gov.in / cybercrime.gov.in/resources).

Used in: Sections 01, 04, Statistics Block

[2] ***Reserve Bank of India — Annual Report 2023-24***

The RBI Annual Report 2023-24 is the flagship publication of the Reserve Bank of India covering monetary policy, financial system stability, and payment systems regulation. The chapter on Payment and Settlement Systems includes a dedicated section on digital payment fraud trends, using data from scheduled commercial banks under the RBI's fraud reporting framework (FMR-1/FMR-2 filings). The 2023-24 edition identifies UPI fraud as the dominant category of digital payment fraud, accounting for approximately 67% of all incidents. The report also discusses RBI's Payments Vision 2025 and regulatory responses including enhanced PIN authentication, velocity controls, and mandatory fraud awareness campaigns. Available at rbi.org.in/Scripts/AnnualReportPublications.aspx.

Used in: Sections 01, 04, Statistics Block

[3] ***National Payments Corporation of India (NPCI) — UPI Product Statistics 2023-24***

The NPCI is the operator and regulator of UPI. Its publicly available product statistics portal provides monthly transaction volume and value data, registered user counts, and participating bank figures. As of early 2024, NPCI data confirmed UPI was processing over 13 billion transactions per month with a total monthly value exceeding Rs. 19 lakh crore. These figures contextualise the scale of UPI adoption and explain why it has become the primary battleground for digital payment fraud. The data also informs the description of UPI's real-time and irrevocable settlement design, which is structurally central to understanding why fraud recovery is so difficult. Available at npci.org.in/what-we-do/upi/product-statistics.

Used in: Section 01

[4] ***Information Technology Act, 2000 — Sections 66C and 66D (Amended 2008)***

The IT Act 2000, as amended by the IT (Amendment) Act 2008, is the primary legislation governing cybercrimes in India. Section 66C defines identity theft as the fraudulent or dishonest use of any person's electronic signature, password, or unique identification feature, and prescribes imprisonment up to three years and a fine up to Rs. 1 lakh. Section 66D defines cheating by personation using computer resources — the specific provision most applicable to UPI fraud scenarios where attackers impersonate bank officials or corporate entities via phone calls and digital interfaces. The full text is available on the Ministry of Electronics and Information Technology (MeitY) portal and the India Code legal database (indiacode.nic.in).

Used in: Section 04 (Legal Consequences)

[5] ***Indian Penal Code, 1860 — Sections 406 and 420***

The Indian Penal Code (IPC) remains applicable alongside the IT Act for financial crimes involving deception and breach of trust. Section 420 (Cheating and dishonestly inducing delivery of property) is the most commonly invoked IPC section in UPI fraud FIRs, with a maximum sentence of seven years imprisonment and a fine. Section 406 (Criminal breach of trust) is applied where the fraudster has been entrusted with money or property, and is particularly relevant in investment and business fraud variants. Both sections form the foundation of most cybercrime FIRs registered at state police stations. Full text available at indiacode.nic.in (Indian Penal Code, 45 of 1860).

Used in: Section 04 (Legal Consequences)

[6] ***RBI Master Direction on Fraud Risk Management in Commercial Banks, 2024***

Released in July 2024, this Master Direction replaced and consolidated all previous RBI circulars on bank fraud reporting and fraud risk management. It introduces updated obligations for banks regarding mule account detection, customer grievance handling for fraud complaints, real-time transaction monitoring thresholds, and inter-bank cooperation for tracing fraud proceeds. Particularly relevant to this case study is the direction's guidance on time-bound freezing of accounts flagged via cybercrime portals and regulatory consequences for non-compliant banks, including reputational risk disclosures and supervisory action. Available at rbi.org.in under Master Directions — Fraud Risk Management in Commercial Banks and Select Financial Institutions.

Used in: Section 04 (Systemic Consequences)